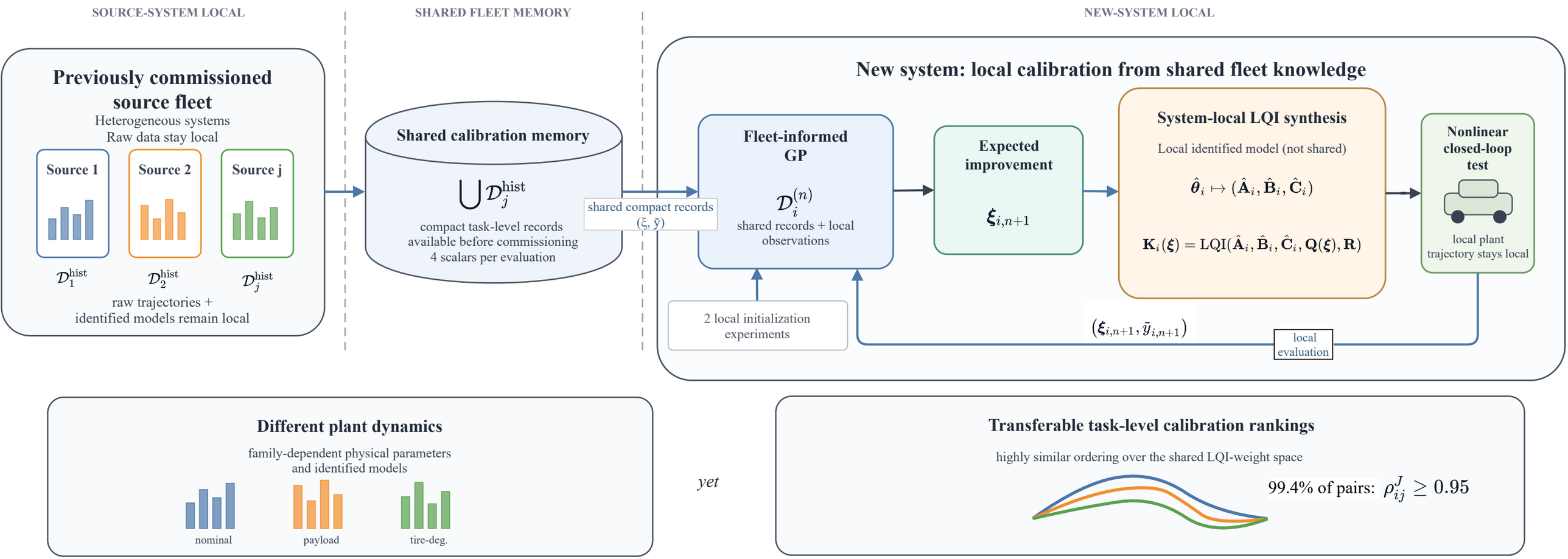

SOURCE-SYSTEM LOCAL
SHARED FLEET MEMORY
NEW-SYSTEM LOCAL
Previously commissioned source fleet
Heterogeneous systems
Raw data stay local
Source 1
Source 2
Source j
$\mathcal{D}_1^{\text{hist}}$
$\mathcal{D}_2^{\text{hist}}$
$\mathcal{D}_j^{\text{hist}}$
raw trajectories + identified models remain local
Shared calibration memory
$\bigcup \mathcal{D}_j^{\text{hist}}$
compact task-level records available before commissioning
4 scalars per evaluation
shared compact records $(\xi, \tilde{y})$
New system: local calibration from shared fleet knowledge
Fleet-informed GP
$\mathcal{D}_i^{(n)}$
shared records + local observations
2 local initialization experiments
Expected improvement
$\xi_{i,n+1}$
System-local LQI synthesis
Local identified model (not shared)
$\hat{\theta}_i \mapsto (\hat{\mathbf{A}}_i, \hat{\mathbf{B}}_i, \hat{\mathbf{C}}_i)$
$\mathbf{K}_i(\xi) = \mathrm{LQI}(\hat{\mathbf{A}}_i, \hat{\mathbf{B}}_i, \hat{\mathbf{C}}_i, \mathbf{Q}(\xi), \mathbf{R})$
Nonlinear closed-loop test
local plant trajectory stays local
$(\xi_{i,n+1}, \tilde{y}_{i,n+1})$
local evaluation
Different plant dynamics
family-dependent physical parameters and identified models
nominal
payload
tire-deg.
yet
Transferable task-level calibration rankings
highly similar ordering over the shared LQI-weight space
99.4% of pairs: $\rho_{ij}^J \geq 0.95$